

Experiment-4A

Aim: Write a Program for session management using cookies and sessions.

Description:

Session management is used in web applications to maintain user information across multiple web pages because HTTP treats each request separately. It helps the website remember users after login and track their activities during a session.

Cookies and sessions are commonly used for this purpose. A cookie stores a small amount of data, such as a session ID, in the user's browser, while the session stores user data securely on the server. When a user accesses the website, the cookie sends the session ID to the server, allowing the application to identify the user and maintain a continuous interaction.

Program:

```
const express = require('express');
const session = require('express-session');
const cookieParser = require('cookie-parser');
const app = express();
const PORT = 5000;
// ===== Middleware =====
app.use(express.json());
app.use(cookieParser());
app.use(
  session({
    secret: 'mySecretKey', // used to sign session ID cookie
    resave: false,
    saveUninitialized: true,
    cookie: { maxAge: 60000 } // session expires in 1 minute
  })
);
// ===== Routes =====
// Home route
app.get('/', (req, res) => {
  res.send('Welcome to Session Management using Express.js');
});
// Login route
app.get('/login', (req, res) => {
  req.session.username = 'Sai Teja';
  res.send('User logged in. Session created.');
```

```
// Profile route (protected)
app.get('/profile', (req, res) => {
  if (req.session.username) {
    res.send(`Welcome ${req.session.username}`);
  } else {
    res.status(401).send('Please login first');
  }
});

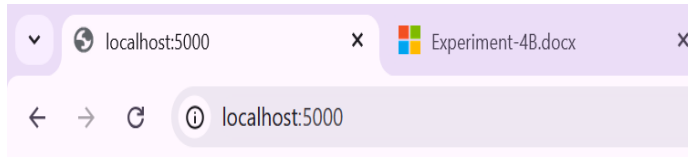
// Count visits using session
app.get('/visit', (req, res) => {
  if (req.session.views) {
    req.session.views++;
  } else {
    req.session.views = 1;
  }
  res.send(`Number of visits: ${req.session.views}`);
});

// Logout route
app.get('/logout', (req, res) => {
  req.session.destroy(err => {
    if (err) {
      return res.send('Error logging out');
    }
    res.clearCookie('connect.sid');
    res.send('Logged out successfully');
  });
});

// ===== Start Server =====
app.listen(PORT, () => {
  console.log(`Server running on http://localhost:${PORT}`);
});
```

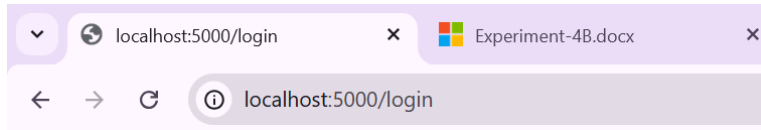
Output:

Step 1: Visit <http://localhost:5000/>



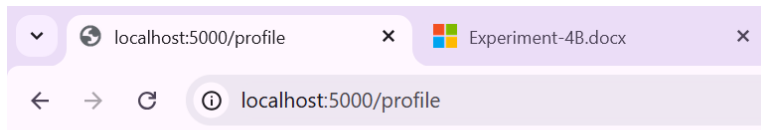
Welcome to Session Management using Express.js

Step 2: <http://localhost:5000/login>



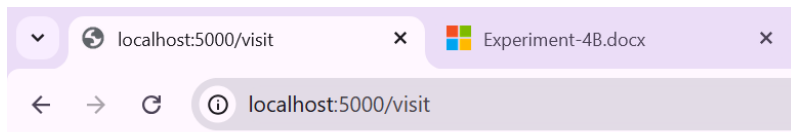
User logged in. Session created.

Step 3: Visit: <http://localhost:5000/profile>



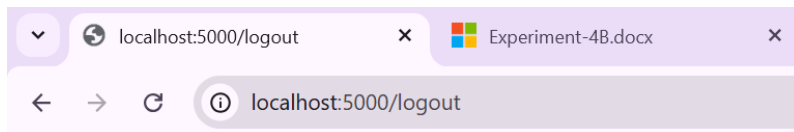
Welcome Sai Teja

Step 4: Visit: <http://localhost:5000/visit>



Number of visits: 1

Step 5: Visit: <http://localhost:5000/logout>



Logged out successfully